

FIN302

Tutorial 8 Answers

Chapter 3

3.14

The formula for the number of contracts that should be shorted gives

$$1.2 \times \frac{20,000,000}{1,080 \times 250} = 88.9$$

Rounding to the nearest whole number, 89 contracts should be shorted. To reduce the beta to 0.6, half of this position, or a short position in 44 contracts, is required.

3.17

The basis is the amount by which the spot price exceeds the futures price. A short hedger is long the asset and short futures contracts. The value of his or her position therefore improves as the basis increases. Similarly, it worsens as the basis decreases.

3.25

A short position in

$$1.3 \times \frac{50,000 \times 30}{50 \times 1,500} = 26$$

contracts is required. It will be profitable if the stock outperforms the market in the sense that its return is greater than that predicted by the capital asset pricing model.

3.32

- (a) The minimum variance hedge ratio, h^* , is $0.95 \times 0.43 / 0.40 = 1.02125$.
- (b) The hedger should take a short position.
- (c) The optimal number of contracts ignoring daily settlement is $1.02125 \times 55,000 / 5,000 = 11.23$ (or 11 when rounded to the nearest whole number)
- (d) The optimal number of contracts is $\hat{\rho} \hat{\sigma}_S V_A / (\hat{\sigma}_F V_F)$ where $\hat{\rho}$ is correlation between percentage one-day returns of spot and futures, $\hat{\sigma}_S$ and $\hat{\sigma}_F$ are the standard deviations of percentage one-day returns on spot and futures, V_A is the value of the position and V_F is the futures price times the size of one contract. In this case, $V_A = 55,000 \times 28 = 1,540,000$ and $V_F = 5,000 \times 27 = 135,000$. If we assume that $\hat{\rho} = 0.95$ and $\hat{\sigma}_S / \hat{\sigma}_F = 0.43 / 0.40 = 1.075$, the optimal number of contracts when daily settlement is considered $0.95 \times 1.075 \times 1,540,000 / 135,000 = 11.65$ (or 12 when rounded to the nearest whole number). This does not make the interest rate adjustment discussed in the final part of Section 3.4.

3.35

- a) The company should short

$$\frac{(1.2 - 0.5) \times 100,000,000}{2,000 \times 250}$$

or 140 contracts.

- b) The company should take a long position in

$$\frac{(1.5 - 1.2) \times 100,000,000}{2,000 \times 250}$$

or 60 contracts.